Ch 15 - Constrained Optimization

Case 1: objective function and constraints are linear.
(Linear programming)

Case 2: Nonlinear constrained optimization.

Linear Programming

objective function:

$$\text{maximize} \quad z = c_1 x_1 + c_2 x_2 + \cdots + c_n x_n$$

(z: profit)

c_j: payoff of each unit of the jth activity that's undertaken

x_j: magnitude of jth activity.

z: total payoff due to n activities.

Constraints $\quad a_{i1} x_1 + a_{i2} x_2 + \cdots + a_{in} x_n \leq b_i$

a_{ij}: amount of the ith resource that is consumed for each unit of jth activity.

b_i: amount of ith resource that is available (The resources are limited)

constaint: $\quad x_i \geq 0$

(e.g. we can't produce negative goods)

Goal: Maximize profit for a number of activities under the constraints.

Ex Set up LP problem

Resource	Product: Regular grade	Product: Premium Grade	Resource Availability
Raw Gas	$7\ m^3$/tonne	$11\ m^3$/tonne	$77\ m^3$/week
Production Time	10 hr/tonne	8hr/ton	80 hr/week
Storage	9 ton	6 ton	
Profit	150 /ton	175 ton	

- only one of the grades can be produced at a time
- Both grades are guaranteed to sell.

Develop a LP formulation to maximize the profit for this operation.

Soln: How much of each gas should be produced to maximize the profit?

Total profit: $150x_1 + 175x_2 = z$

maximize $z = 150x_1 + 175x_2$: objective function

Total gas used $\begin{cases} 7x_1 + 11x_2 \leq 77 & \text{constraint (raw material)} \\ 10x_1 + 8x_2 \leq 80 & \text{time constraint} \\ x_1 \leq 9 & \text{regular storage constraint} \\ x_2 \leq 6 & \text{premium} \quad " \quad " \\ x_1, x_2 \geq 0 & \text{positivity constraints} \end{cases}$

constraints

Graphical Solutions

(Limited to 2D or 3D)

maximize $z = 150x_1 + 175x_2$

subject to:

① $7x_1 + 11x_2 \leq 77 \Rightarrow x_2 = -\frac{7}{11}x_1 + 7$

② $10x_1 + 8x_2 \leq 80$

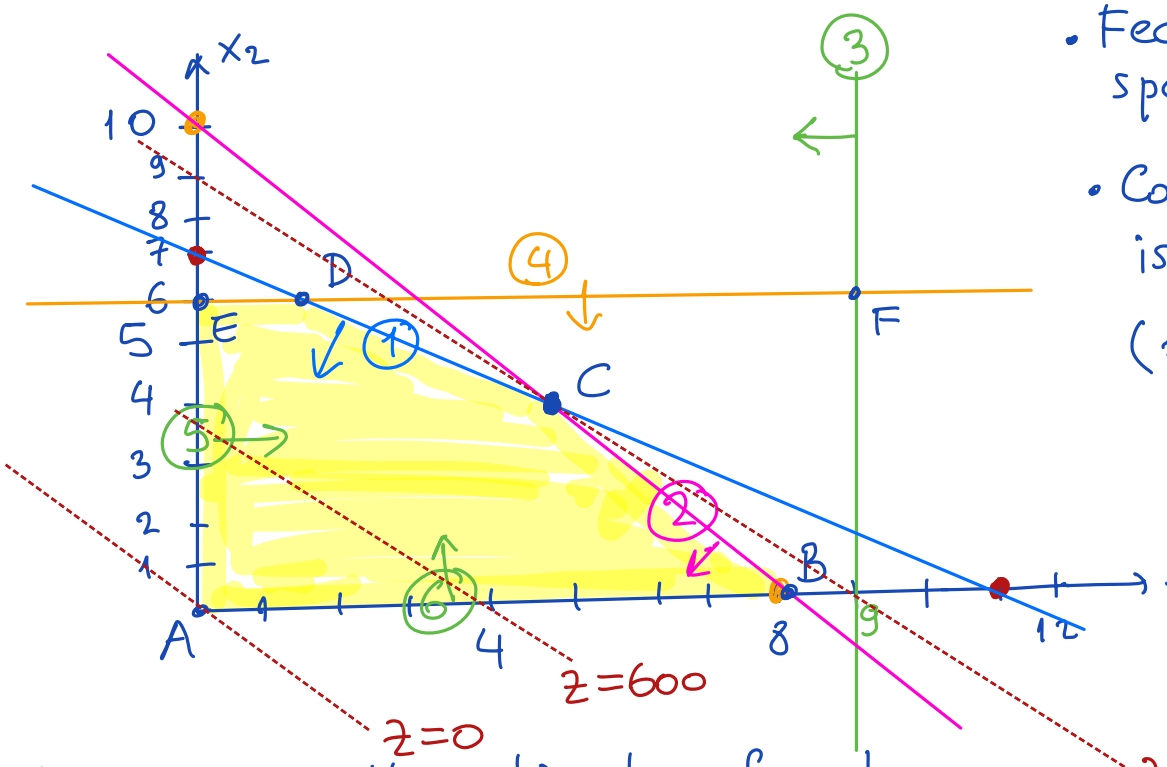
③ $x_1 \leq 9$

④ $x_2 \leq 6$

⑤ $x_1 \geq 0$

⑥ $x_2 \geq 0$

Step 1: Draw the constraints:



- Feasible solution space: ABCDE
- Constraint ③ is redundant

(Feasible solution space will not be affected if we remove it.)

Step 2: Draw the objective function

$z = 150x_1 + 175x_2$

at $z = 0 = 150x_1 + 175x_2 \Rightarrow x_2 = -\frac{150}{175}x_1$

at $z = 600 = 150x_1 + 175x_2 \Rightarrow x_2 = \frac{-600}{175} - \frac{150}{175}x_1$

2pts: (0, 3.42), (4, 0)

increase $z = 1400 = 150x_1 + 175x_2$

at $\boxed{x_1 = 4.9 \;, x_2 = 3.9}$ max profit of 1400

Substitute the answer(s) into the constraints

① $7(4.9)+11(3.9) \cong 77$
② $10(4.9)+8(3.9) \cong 80$

} just meet the resource and time constraints (binding constraints)

$4.9 \leq 9$
$3.9 \leq 6$

} non-binding constraints

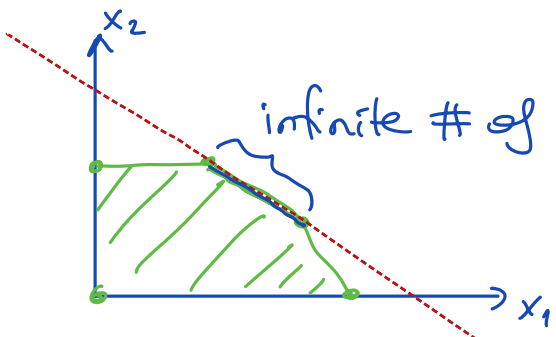
Conclusions:

- we can increase our profit by increasing resources: the raw gas or production time (binding constraints)
- increasing storage will not increase the profit

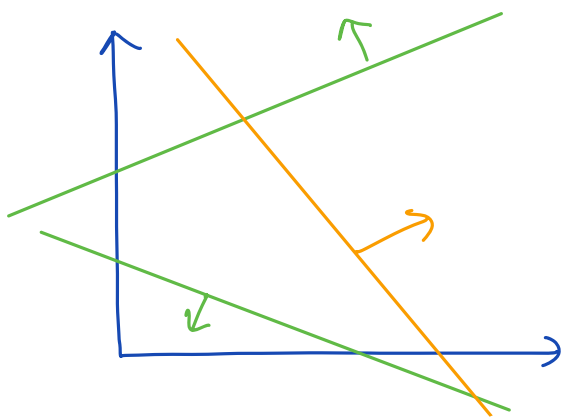
Four possible outcomes in a LP problem:

1) Unique solution: max. of the objective function intersects at a single point. (prev. example)

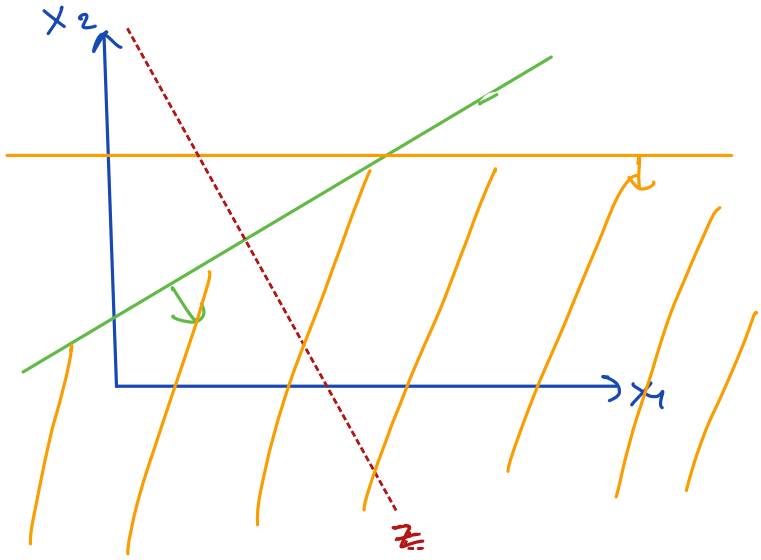
2) Alternate solutions: objective function is parallel to one of the constraints



3) No feasible solution: may happen if problem is overconstrained



4) Unbounded problems: problem is underconstrained



If you have a unique soln:

- optimum is at one of the corner ptr where two constraints meet ⇒ extreme pt
- Focus on extreme pts to narrow down the options.
- Not all extreme pts are feasible (point F in figure)
- Narrow down soln feasible extreme ptrs.

MATCAB Functions for Optimization

fminbnd: min. of a single variable function on a fixed (constrained) interval

$$\min_x f(x) \quad \text{such that} \quad x_1 < x < x_2$$

Syntax:

[x,fval] = fminbnd(fun, x1, x2, options)

EX Find the max. of $f(x) = 2\sin x - \frac{x^2}{2}$ within interval $x_l = 0$ and $x_u = 4$

save in file fx.m

```
function f = fx(x)
f = -(2*sin(x) - x^2/2);
```

minimizing $-f(x)$ is equal to maximizing $f(x)$

```
>> x = fminbnd('fx', 0, 4)
```

Result:

$f = -1.7757$

$x = 1.4275$

fminsearch: minimize a function of several variable (unconstrained)

$$\min_{\underline{x}} f(\underline{x})$$

Syntax:

[x,fval] = fminsearch(fun, x0, options) → x0: initial guess

EX

maximize $f(x_1, x_2) = 2 + x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$

initial guess $x_1 = 0.5$, $x_2 = 0.5$

```
>> f = @(x) -(2 + x(1) - x(2) + 2*x(1)^2 + 2*x(1)*x(2) + x(2)^2);
>> [x,fval] = fminsearch(f, [-0.5, 0.5])
```

$x = [-1.0 \quad 1.5]$

fval : 0.75

fminunc : (unconstrained) Min. of unconstrained multivariable function

$$\min_{\underline{x}} f(\underline{x})$$

Syntax : `[x, fval] = fminunc(fun, x0, options)`

fmincon : (constrained) Find the min of constrained multivariable function

$$\min_{\underline{x}} f(\underline{x}) \quad \text{such that} \quad \begin{cases} c(\underline{x}) \leq 0 \\ ceq(\underline{x}) = 0 \\ A\underline{x} \leq b \\ A_{eq}\, x = b_{eq} \\ lb \leq x \leq ub \end{cases}$$